

HFT Ops Interview Preparation

Probability / Brainteasers / Code

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1 Math / Probability Theory

1.1 Combinatorics

1.1.1 Permutations

Given a problem with n total options, of which we want to choose k in a particular order, we can compute the solution as:

$$P(n, k) = \frac{n!}{(n - k)!}$$

Suppose a different situation occurs where there are repeats among the elements. If some elements are identical, we divide by the factorial of each element's frequency to avoid counting duplicate arrangements:

$$P_{\text{repetitions}} = \frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

1.1.2 Combinations

For cases where the order among the selected elements *does not* matter then we compute the possibilities as:

$$C(n, k) = \frac{n!}{k! \times (n - k)!}$$

If we substitute in the equation for permutations, we can see that this is equivalent to the number of permutations, dividing out the cases of reordering the elements:

$$C(n, k) = \frac{P(n, k)}{k!}$$

1.1.3 Distributing Indistinguishable Objects

We model this as arranging n identical items (stars) and $k - 1$ dividers (bars) in a line. Each arrangement partitions the items into k bins. The problem reduces to choosing $k - 1$ positions for dividers among $n + (k - 1)$ total positions:

$$C(n + k - 1, k - 1) = \binom{n + k - 1}{k - 1}$$

1.1.4 Inclusion-Exclusion

When counting the number of unique elements (cardinality) in a union of sets, we find the sum by counting each set individually, then subtracting the overlap:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Which extends to a larger number of sets as:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Note that after removing the overlap from each pairwise intersection, we had to add BACK in the triple intersection, which would have been subtracted too many times. This pattern follows for higher orders of sets, where the odd length items (singles, triples) are added and the even length items (doubles, quadruples) are subtracted.

1.1.5 Select Worked Problems

Problem 1.1

How many ways can you arrange the letters in the word BANANA?

Solution. There are $n = 6$ elements. Repeated items have frequency 3 and 2 (Ns and As). Then the number of arrangements are $\frac{6!}{3! \times 2!} = \frac{720}{12} = 60$.

Problem 1.2

You roll 2 dice. How many distinct outcomes are there? What if you can't tell the dice apart?

Solution. With two dice, there are $6 \times 6 = 36$ outcomes if we can tell the dice apart. If we *can't* tell the dice apart, then we can solve either logically or formulaically:

Logically, we can see that from the original 36 combinations, there are 6 outcomes where the dice are the same, leaving 30 outcomes using mixed pairs of dice. Of these, half are duplicate when we can't tell the dice apart, leaving 15 outcomes. We can then add the 6 double outcomes back in for a total of 21 outcomes.

Formulaically, we can solve using a similar approach, taking the sum of mixed pair outcomes and double outcomes as $C(6, 2) + C(6, 1) = 15 + 6 = 21$.

Lastly, we can consider this problem using the stars and bars approach, where we notice that there are $n = 2$ identical dice and 6 distinct bins with $k = 5$ dividers between them. Then we can count the outcomes by $C(2 + 6 - 1, 6 - 1) = C(7, 5) = 21$

Problem 1.3

How many ways can you distribute 8 identical balls into 3 distinct boxes?

Solution. This is a direct application of stars and bars with $n = 8$ and $k = 3$. The number of outcomes is $C(8 + 3 - 1, 3 - 1) = C(10, 2) = \frac{10 \times 9 \times 8 \times \dots \times 3}{2} = 45$.

Problem 1.4

A deck has 52 cards. How many 5-card hands contain at least one ace?

Solution. Since this solution asks for the number of hands containing at least one ace, it will be simpler to compute the complement. This is the number of hands containing no aces, we then subtract from the total number of 5 card hands: $C(52, 5) - C(48, 5) = 886,656$.

Problem 1.5

You have 5 books on a shelf. In how many ways can you choose 3 books and arrange them left-to-right on a new shelf?

Solution. This is simply $C(5, 3) \times P(3, 3) = \frac{5 \times 4}{3!} \times \frac{3!}{0!} = \frac{60 \times 3!}{3!} = 60$.

Problem 1.6

A pizza place offers 8 toppings. How many different pizzas can you make if you can choose any number of toppings (including none)?

Solution. We can solve this by taking the summation of $C(8, i)$ from $i = 0$ to $i = 8$:

$$Total\ outcomes = \sum_{i=0}^8 C(8, i) = 256$$

There is a more succinct solution we can find if we approach the problem seeing each topping as the binary choice of being either on or off of the pizza, then the total outcomes are $2^8 = 256$.

Problem 1.7

How many ways can 12 people be split into 3 groups of 4?

Solution. We choose each group independently with the remaining members of the pool, and then divide by the reorderings since the groups are indistinguishable:

$$\frac{C(12, 4) \times C(8, 4) \times C(4, 4)}{3!} = \frac{495 \times 70 \times 1}{6} = 34,650.$$

1.2 Probability

Definition 1.1 (Sample Space). The sample space is the set of all possible outcomes of an event.

Many probability problems require defining the sample as an initial step, often followed by identifying the favourable outcomes.

Definition 1.2 (Events). The subset of the sample space containing the outcomes of interest.

1.2.1 Complement

Sometimes, probability-based questions which require computing many cases can be solved by instead calculating the complementary case using the identity

$$P(A) = 1 - P(\neg A)$$

Such as the previous “at least one...” scenario, where we may need to calculate several individual subcases if we do not use the complement.

1.2.2 Independence

If the outcomes of two events do *not* affect each other, then the two events are independent. Formally:

$$P(A \cap B) = P(A) \times P(B)$$

A common occurrence is when choosing elements from a group. If we choose items *with* replacement, then each draw is independent. Without replacement the pool changes size and composition after each draw.

1.2.3 Mutual Exclusivity

On the other end, there are events that cannot both happen at the same time, then:

$$P(A \cup B) = P(A) + P(B)$$

We can note that this is really just an application of the inclusion-exclusion equation, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where the intersection is zero.

1.2.4 Select Worked Problems

Problem 1.8

How many people do you need in a room for a $>50\%$ chance that two share a birthday?

Solution. Computing this directly is impractical, since there are many ways for multiple people to share a birthday. Instead, using the complement, we simply subtract the probability that *no* people share a birthday from 1:

$$P(\text{all different}) = \prod_{i=1}^n \frac{365 - i + 1}{365}$$

We multiply additional factors until the quantity drops *below* 0.5 (because then $1 - P$ is greater than 0.5). This turns out to be $n = 23$, where $P(\text{all different}) \approx 0.493$ and thus $P(\text{birthday is shared}) \approx 0.507$.

Problem 1.9

You flip a fair coin 3 times. What's the probability of getting exactly 2 heads?

Solution. Flipping a coin three times has eight outcomes. Of these, three involve exactly two heads, thus the probability is $\frac{3}{8} = 0.375$.

If we don't want to enumerate each of the eight options and count directly, we could use $C(3, 2) = 3$ to find the number of outcomes involving two heads.

Problem 1.10

Roll two dice. What's the probability the sum is 7?

Solution. There are 36 outcomes of rolling two dice. Of these, the pairs that equal 7 are $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$. Then the probability is $\frac{6}{36} \approx 0.167$.

Problem 1.11

You draw 2 cards from a standard 52-card deck (without replacement). What's the probability both are hearts?

Solution. Without replacement, the draws are dependent. Then, the probability is:

$$\frac{13}{52} \times \frac{12}{51} \approx 0.0588$$

Note that *both* the numerator and the denominator shrank after we removed the first card.

We could also solve this by counting, if we divide the number of ways to pick two hearts by the total number of ways to pick two cards:

$$\frac{C(13, 2)}{C(52, 2)} = \frac{78}{1326} \approx 0.0588$$

Problem 1.12

You flip a fair coin 5 times. What's the probability of getting at least one head?

Solution. We can use the complement, the probability of getting *no* heads, subtracted from 1:

$$1 - \frac{1}{2^5} = 0.96875$$

Problem 1.13

You roll a fair die twice. What's the probability the second roll is strictly greater than the first?

Solution. There are 36 total ways to roll two dice. If the first roll is a 1, then there are 5 ways to roll higher on the second die. Similarly, there are 4 ways to roll higher if we rolled a 2, and so on.

Thus the number of favourable outcomes $1 + 2 + \cdots + 5$ or the triangle number of 5, $T_5 = 15$. The final answer is:

$$\frac{15}{36} \approx 0.417$$

Alternatively, we could solve by symmetry, calling roll 1 and 2 as A and B , respectively. If we notice that $P(A > B) = P(B < A)$, and $P(A = B) = \frac{6}{36}$, then we evenly divide the remaining outcomes by 2 to get $\frac{15}{36}$.

Problem 1.14

A box has 5 white and 3 black balls. You draw 3 without replacement. What's the probability of drawing exactly 2 white and 1 black?

Solution. The total ways to draw 3 balls are $C(8, 3) = 56$. We then find the ways to choose 2 white balls and 1 black ball individually, which is $C(5, 2)$ and $C(3, 1)$, then the answer is:

$$\frac{C(5, 2) \times C(3, 1)}{C(8, 3)} = \frac{10 \times 3}{56} \approx 0.536$$

Since we remove the balls without replacement, this problem follows the hypergeometric distribution.

Problem 1.15

You randomly pick a number from 1 to 100. What's the probability it's divisible by 3 or 5?

Solution. There are 33 numbers divisible by 3, and 20 numbers divisible by 5. The numbers divisible by both are the multiples of 15 of which there are 6 (15, 30, 45, 60, 75, 90). Thus:

$$P_{Div}(A \cup B) = \frac{33}{100} + \frac{20}{100} - \frac{6}{100} = 0.47$$

Problem 1.16

A family has 3 children. Assuming each child is equally likely to be a boy or girl, what's the probability that all 3 are the same gender?

Solution. There are eight outcomes, of those there are two in which all children are the same gender, so the answer is $\frac{2}{8} = 0.25$.

Problem 1.17

You throw 3 dice. What's the probability ALL three show different numbers?

Solution. There are $6^3 = 216$ total ways to throw three dice. After we throw the first dice, there are 5 options remaining, and 4 after the second dice. Then the answer is:

$$\frac{6 \times 5 \times 4}{6^3} = \frac{120}{216} \approx 0.556$$

Problem 1.18

You toss a fair coin until you get heads. What's the probability you need exactly 3 tosses?

Solution. We stop at the first heads, so needing exactly 3 tosses means the first two tosses are tails and the third is heads: T, T, H. Since tosses are independent and each outcome has probability $\frac{1}{2}$,

$$P(\text{exactly 3}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}.$$

We do not enumerate anything else, because exactly one sequence produces this event.

Remark (Two ways probability gets counted). The reason we can pin down this single case without listing every outcome is that this problem lives in a different kind of sample space than counting problems like the birthday question.

When every outcome is equally likely, probability is combinatorial. You divide favorable outcomes by total outcomes, so you have to characterize the entire sample space. Each outcome carries no probability on its own and earns weight only by being counted against the total.

When outcomes are not equally likely, each outcome carries its own probability directly, usually a product of independent step probabilities. Here the sequence TTH has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ by itself, with no denominator to divide by. That is why one case can be solved in isolation.

These are the same axioms applied to different spaces. Counting favorable over total is the special case that appears when all per-outcome probabilities are equal, so the sum of probabilities collapses into a plain count. The coin problem is the general version, where the per-outcome probabilities differ and you take the one you want.

Problem 1.19

In a room of 5 people, what's the probability that at least two share a birthday? (Assume 365 days, equally likely.)

Solution. Again, we use the complement to find 1 minus the probability that no one shares a birthday:

$$1 - \left(\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \frac{362}{365} \times \frac{361}{365} \right) \approx 1 - 0.9729 \approx 0.0271$$

1.3 Expected Value

Definition 1.3 (Expected Value). The long-run average outcome of a random process:

$$\mathbb{E}[X] = \sum x_i \times P(x_i)$$

Which is simply the weighted average of all possible outcomes. A note is that the EV is neither required to be a possible outcome (e.g., the EV of a dice roll is 3.5) nor the most likely outcome.

Theorem 1.4 (Linearity of Expectation). *The EV of multiple events is the sum of their EVs:*

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

This holds always, even if X and Y are dependent.

Definition 1.5 (Indicator Random Variables). A variable I_A that equals 1 when event A occurs and 0 otherwise. Its expectation is just the event's probability, since $\mathbb{E}[I_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$.

The point is not the variable itself, which is trivial, but the decomposition it enables. To find the expected *count* of something, write that count as a sum of indicators, one per event, then apply linearity:

$$\mathbb{E} \left[\sum_i I_{A_i} \right] = \sum_i P(A_i).$$

This turns a hard count with an ugly distribution into a sum of easy probabilities, and because linearity ignores dependence, it works even when the events are entangled.

Theorem 1.6 (Law of Total Expectation). *For any partition of the sample space into disjoint cases $B_1, \dots, B_n$ with $\sum_i P(B_i) = 1$,*

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X \mid B_i] \cdot P(B_i).$$

This is the definition of expected value regrouped by cases: instead of averaging over individual outcomes, average the conditional mean within each case, weighted by how likely that case is. Mathematically it adds nothing new, since expanding each $\mathbb{E}[X \mid B_i]$ recovers the flat sum. Operationally it is a different tool, useful in three situations:

- *when you are handed conditional averages and case probabilities rather than the outcome distribution*
- *when X is continuous, so the flat sum $\sum_x x P(X = x)$ does not apply but conditioning on discrete cases still does;*

- when $\mathbb{E}[X \mid B_i]$ refers back to $\mathbb{E}[X]$ itself, turning the conditioning into a solvable equation

The cases must be mutually exclusive and cover everything. Overlapping or incomplete cases break the weighting.

Example 1.7 (Conditioning on cases). A commute depends on weather. Sunny days are 60% of days with a 20 minute average; rainy days are 40% with a 35 minute average. The two cases partition every day, so

$$\mathbb{E}[\text{commute}] = 0.6(20) + 0.4(35) = 12 + 14 = 26 \text{ minutes.}$$

Example 1.8 (Recursive EV). Suppose we were interested in the expected number of rolls we would take if we continue rolling a die until we hit a 6. If we call E the expected number of rolls, then:

$$E = 1 + \frac{5}{6} \cdot E \longrightarrow E - \frac{5}{6} \cdot E = 1 \longrightarrow E = 6$$

We can note that this is the EV of the geometric distribution, the expected trials until first success with probability $\frac{1}{p}$.

1.3.1 Select Worked Problems

Problem 1.20

A game costs \$1. You flip a coin: heads you win \$3, tails you win nothing. What's $\mathbb{E}[\text{Profit}]$?

Solution. $EV = 0.5 \cdot 2 + 0.5 \cdot -1 = +0.5$

Problem 1.21

Roll a die. If it shows 6, you roll again and add. Otherwise, your score is the number shown. What's the expected score?

Solution.

$$EV = 3.5 + \frac{1}{6} \cdot EV \longrightarrow \frac{5}{6} \cdot EV = 3.5 \longrightarrow EV = 4.2$$

Problem 1.22

Shuffle a standard deck. What's the expected number of cards that are in their "correct" position (card k is in position k)?

Solution. This is linearity of expectation applied to indicators. Let $I_k = 1$ if card k lands in position k , and 0 otherwise. In a random shuffle each card is equally likely to occupy any of the 52 positions, so $\mathbb{E}[I_k] = P(I_k = 1) = \frac{1}{52}$. The number of correctly placed cards is $M = \sum_{k=1}^{52} I_k$, and by linearity

$$\mathbb{E}[M] = \sum_{k=1}^{52} \mathbb{E}[I_k] = 52 \cdot \frac{1}{52} = 1.$$

The I_k are dependent, but linearity does not require independence, so that never matters.

Problem 1.23

You flip a coin until you get heads. How many flips do you expect?

Solution.

$$EV = 1 + 0.5 \cdot EV \longrightarrow 0.5 \cdot EV = 1 \longrightarrow EV = 2$$

Problem 1.24

A cereal box contains 1 of 4 different toys (equally likely). How many boxes must you buy to expect to collect all 4?

Solution. The total number of boxes is a sum of four phase waits, so we use linearity of expectation. In phase k we already hold $k - 1$ toys and buy boxes until a new one appears. The per-trial success probability is $p = \frac{\text{favourable}}{\text{total}}$, the number of missing toys over 4. We want the reciprocal, the expected number of trials per success, which is $\frac{1}{p}$:

$$\begin{aligned} 0 \text{ toys, } p &= \frac{4}{4} \longrightarrow \mathbb{E}[T_1] = \frac{1}{p} = 1 \\ 1 \text{ toy, } p &= \frac{3}{4} \longrightarrow \mathbb{E}[T_2] = \frac{1}{p} = \frac{4}{3} \\ 2 \text{ toys, } p &= \frac{2}{4} \longrightarrow \mathbb{E}[T_3] = \frac{1}{p} = 2 \\ 3 \text{ toys, } p &= \frac{1}{4} \longrightarrow \mathbb{E}[T_4] = \frac{1}{p} = 4 \end{aligned}$$

$$\mathbb{E}[T] = \mathbb{E}[T_1] + \mathbb{E}[T_2] + \mathbb{E}[T_3] + \mathbb{E}[T_4] = 1 + \frac{4}{3} + 2 + 4 = \frac{25}{3} \approx 8.33$$

Problem 1.25

I roll a die repeatedly and keep a running sum. What's the expected number of rolls until the sum first reaches or exceeds 10?

Solution. We solve recursively. Let $\mathbb{E}[s]$ be the expected number of additional rolls needed to first reach a sum of at least 10, starting from current sum s . The answer we want is $\mathbb{E}[0]$.

Base case: $\mathbb{E}[s] = 0$ for all $s \geq 10$, since the target is already met. For $s < 10$, one roll costs 1 and moves to $s + 1, \dots, s + 6$ with equal probability $\frac{1}{6}$:

$$\mathbb{E}[s] = 1 + \frac{1}{6} \sum_{j=1}^6 \mathbb{E}[s + j], \quad \mathbb{E}[s] = 0 \text{ for } s \geq 10.$$

Working downward from the base case:

$$\begin{aligned} \mathbb{E}[9] &= 1 + \frac{1}{6}(\mathbb{E}[10] + \dots + \mathbb{E}[15]) = 1 \\ \mathbb{E}[8] &= 1 + \frac{1}{6}\mathbb{E}[9] = \frac{7}{6} \\ \mathbb{E}[7] &= 1 + \frac{1}{6}(\mathbb{E}[8] + \mathbb{E}[9]) = \frac{49}{36} \\ &\vdots \\ \mathbb{E}[0] &\approx 3.6 \end{aligned}$$

Problem 1.26

You roll a fair six-sided die. After seeing the result, you may keep it as your score, or reroll. If you reroll, the new value replaces the old one (you cannot go back to a discarded roll). You are allowed up to three rolls total, and your score is the value you are holding when you stop. Playing optimally to maximize your expected score, what is that expected score, and what is the stopping rule at each stage?

Solution. Let $\mathbb{E}[r]$ be the expected score with r rerolls remaining. Solve backwards.

With no rerolls left, you keep whatever you roll, so $\mathbb{E}[0] = 3.5$.

With one reroll left, you keep the current face only if it beats the continuation value $\mathbb{E}[0] = 3.5$, i.e. keep on $\{4, 5, 6\}$ and reroll otherwise:

$$\mathbb{E}[1] = \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6) + \frac{3}{6}(3.5) = \frac{17}{4} = 4.25.$$

With two rerolls left, the continuation value is now $\mathbb{E}[1] = 4.25$, so you keep only faces that beat it, $\{5, 6\}$, and reroll on $\{1, 2, 3, 4\}$:

$$\mathbb{E}[2] = \frac{1}{6}(6) + \frac{1}{6}(5) + \frac{4}{6}(4.25) = \frac{14}{3} \approx 4.67.$$

Problem 1.27

You pick a random number uniformly from $[0, 1]$. Then you pick another. Then another. You stop when the running total exceeds 1. How many numbers do you expect to pick?

Solution. Let N be the number of draws until the running sum first exceeds 1, and let $X_1, X_2, \dots$ be the draws. We want $\mathbb{E}[N]$.

Step 1: a tail probability. The *tail probability* $\mathbb{P}(N > n)$ is the chance N exceeds a cutoff n , i.e. the chance we still need more draws after n of them. Needing more than n draws means the first n have not yet crossed 1:

$$N > n \iff X_1 + \dots + X_n \leq 1.$$

Step 2: geometry The point $(X_1, \dots, X_n)$ lands uniformly in the unit hypercube $[0, 1]^n$, which has volume 1. Since the draws are uniform, the probability of landing in any region equals that region's volume. The region where the coordinates are nonnegative and sum to at most 1 is a *simplex* (the n -dimensional triangle: a segment in 1D, triangle in 2D, tetrahedron in 3D). Its volume is $\frac{1}{n!}$ ($V_n(t) = \int_0^t V_{n-1}(t-x) dx$, starting from $V_1(t) = t$. Then $V_n(t) = \frac{t^n}{n!}$, evaluated at $t = 1$, $V_n(1) = \frac{1}{n!}$), so

$$\mathbb{P}(N > n) = \mathbb{P}(X_1 + \dots + X_n \leq 1) = \frac{1}{n!}.$$

Step 3: sum the tails. For a nonnegative integer variable, the mean equals the sum of its tail probabilities:

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \mathbb{P}(N > n).$$

This holds because an outcome $N = k$ clears the cutoffs $n = 0, 1, \dots, k-1$, so it is counted exactly k times across the sum, contributing its own value. Substituting the tails:

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \dots = e \approx 2.718.$$

1.4 Conditional Probability & Bayes

Definition 1.9 (Conditional Probability). The probability of A , given that B has already happened:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Theorem 1.10 (Bayes' Theorem). Baye's theorem allows us to update the probability of a hypothesis as more information becomes available:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- $P(A|B)$ is the posterior probability
- $P(B|A)$ is the likelihood
- $P(A)$ is the prior probability
- $P(B)$ is the marginal probability

Theorem 1.11 (Law of Total Probability). If $B_1, B_2, \dots, B_n$ partition the sample space (are mutually exclusive and exhaustive), then:

$$P(A) = \sum P(A|B_i) \times P(B_i)$$

We often use this to compute $P(B)$ when we don't know it directly, but know both $P(B|A)$ and $P(B|\neg A)$.

Example 1.12 (Monty Hall Problem). Assume the contestant chooses Door 1 and Monty opens Door 2 to reveal a goat. Let A_i be the event the car is behind Door i (prior $P(A_i) = \frac{1}{3}$), and B be the event Monty opens Door 2.

The likelihoods $P(B|A_i)$ are:

- $P(B|A_1) = \frac{1}{2}$ (Monty randomly picks Door 2 or 3)
- $P(B|A_2) = 0$ (Monty cannot reveal the car)
- $P(B|A_3) = 1$ (Monty is forced to open Door 2)

The marginal probability of B is:

$$P(B) = \sum_{i=1}^3 P(B|A_i)P(A_i) = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + (0)\left(\frac{1}{3}\right) + (1)\left(\frac{1}{3}\right) = \frac{1}{2}$$

By Bayes' Theorem, the posterior probabilities of winning by staying (A_1) or switching (A_3) are:

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B)} = \frac{\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}{\frac{1}{2}} = \frac{1/6}{1/2} = \frac{1}{3}$$
$$P(A_3|B) = \frac{P(B|A_3)P(A_3)}{P(B)} = \frac{(1)\left(\frac{1}{3}\right)}{\frac{1}{2}} = \frac{1/3}{1/2} = \frac{2}{3}$$

Switching doubles the probability of winning.

1.4.1 Select Worked Problems

Problem 1.28

I roll two dice. Given that the sum is 8, what's the probability that one of the dice shows a 3?

Solution. The possible ways to roll a sum of 8 are: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2). Of these, two options have a 3 in them, so the answer is $\frac{2}{5} = 0.4$.

Problem 1.29

A bag has 3 red and 2 blue balls. You draw one, don't replace it, then draw another. What's $P(\text{2nd is red} \mid \text{1st was blue})$?

Solution. Once we drew one blue on, the remaining bag composition is 3 red balls and 1 blue one. Then, there is a $\frac{3}{4}$ probability that the second will be red.

Problem 1.30

60% of emails are spam. A filter catches 95% of spam but also flags 3% of legitimate emails. An email is flagged. What's the probability it's actually spam?

Solution. We want $P(\text{Spam} \mid \text{Flagged})$. The likelihood $P(\text{Flagged} \mid \text{Spam})$ is given as 0.95, the prior $P(\text{Spam})$ is 0.6, and the marginal probability $P(\text{Flagged})$ is $0.95 \cdot 0.6 + 0.03 \cdot 0.4 = 0.582$.

Then the posterior is

$$P(\text{Spam} \mid \text{Flagged}) = \frac{P(\text{Flagged} \mid \text{Spam}) \times P(\text{Spam})}{P(\text{Flagged})} = \frac{0.95 \times 0.6}{0.582} \approx 0.979$$

Problem 1.31

You have 2 coins. Coin A is fair (50% heads). Coin B is biased (75% heads). You pick a coin at random and flip it. It lands heads. What's the probability you picked coin B?

Solution. We are looking for the posterior, $P(B \mid H)$.

The priors are $P(A) = P(B) = 0.5$. The likelihoods are $P(H \mid A) = 0.5$ and $P(H \mid B) = 0.75$. Then $P(H) = P(H \mid A) \cdot P(A) + P(H \mid B) \cdot P(B) = 0.5 \cdot 0.5 + 0.75 \cdot 0.5 = 0.625$. Then finally

$$P(B \mid H) = \frac{P(H \mid B) \cdot P(B)}{P(H)} = \frac{0.75 \cdot 0.5}{0.625} = 0.6$$

After flipping and landing on heads, our belief that we picked coin B increased from 0.5 to 0.6.

Problem 1.32

Same as 1.31. The chosen coin is flipped twice and get heads both times. Now what's $P(B)$?

Solution. The only difference is our priors are $P(A) = 0.4$ and $P(B) = 0.6$. Then we update $P(H) = P(H \mid A) \cdot P(A) + P(H \mid B) \cdot P(B) = 0.5 \cdot 0.4 + 0.75 \cdot 0.6 = 0.65$. Finally:

$$P(B \mid H) = \frac{P(H \mid B) \cdot P(B)}{P(H)} = \frac{0.75 \cdot 0.6}{0.65} = \frac{9}{13} \approx 0.692$$

Problem 1.33

A family has 2 children. Given that at least one child is a boy, what's the probability both are boys?

Solution. We can answer this by considering the restriction over the sample space caused by the new information. Originally, for two children there would be four possibilities, of which 1 has two boys. After we know that at least one child is a boy, only three options remain $(B, G), (B, B), (G, B)$. Thus, the probability both are boys is $\frac{1}{3}$.

Problem 1.34

Three cards: one is red on both sides, one is blue on both sides, one is red on one side and blue on the other. You pick a card at random and see a red side. What's the probability the other side is also red?

Solution. A

2 Brainteasers

Problem 2.1 [Easy]

Five pirates need to divide 100 Gold Coins. Pirates have a hierarchy, from Level 1 to level 5. The highest level pirate is the leader. The leader proposes a plan to distribute the gold and all the pirates vote on it (including the leader). If at least 50% of the pirates agree on the plan, the gold is split according to the proposal. If not, level-5 pirate is kicked from the ship, and the level-4 pirate now proposes a new plan. This process continues until a proposal is accepted. All pirates are extremely smart and extremely greedy. How does level-5 Pirate divide the treasure?

Solution. We approach this problem game-theoretically and dynamically:

- 2 pirates → Pirate 2 proposes 100 coins for themselves and zero for Pirate 1
- 3 pirates → Pirate 3 bribes Pirate 1 with one coin and keeps 99 for themselves, which they accept because they get zero coins if Pirate 3 is booted off.
- 4 pirates → Pirate 4 bribes Pirate 2 with one coin and keeps 99 for themselves, in the same manner as the 3-pirate case.
- 5 pirates → Pirate 5 bribes pirates 1 and 3 with one coin each, keeping 98 for themselves.

Thus, the final division is [98, 0, 1, 0, 1].

Problem 2.2 [Easy]

An explorer travels 1 mile South, 1 mile East, then 1 mile North, and finds themselves back at the starting point. The explorer spots a bear, what is the color of the bear?

Solution. The bear must be white. There are a few relevant scenarios that could have resulted in the explorer starting back where they began. In the first case, they started at the North pole, then travelling South equates to travelling directly away from the Pole. They can then travel *any* distance East or West, the following mile back North will return them to their destination. There are no bears in the South Pole so we omit this, though the same logic applies (with the Eastward movement equivalent to spinning in place or not moving at all).

There are other solutions near the South Pole. If the traveler's initial one-mile journey South ends on a latitude circle with a circumference of $\frac{1}{n}$ ($n \in \mathbb{Z}^+$) miles, walking one mile East will complete n full rotations around the pole and leave them exactly where they started walking East, meaning the following mile North would place them in the same starting position.

Problem 2.3 [Easy]

There is a light bulb inside a room and three switches outside. All switches are currently in off state and only one switch controls the light bulb. You may turn any number of switches on or off any number of times you want. How many times do you need to go into the room to figure out which switch controls the light bulb?

Solution. We only need to enter the room once. The first switch is flipped and left on for enough time that the bulb becomes hot. It is then flipped off and the second switch is flipped, we then immediately enter the room. This leads to three cases:

1. Bulb is hot → the light must be off and the first switch controls the light

2. Bulb is on $\rightarrow$ the second switch controls the light
3. Bulb is off and cold $\rightarrow$ the third switch controls the light

If there were four switches, we can still infer the switch in one entry by turning on both switches 1 and 2 for a long time, then switching off switch 2 and turning on switch 3 before entering the room. This results in the following cases:

1. Bulb is on and hot $\rightarrow$ Switch 1 controls the light
2. Bulb is on and cool $\rightarrow$ Switch 3 controls the light
3. Bulb is off and hot $\rightarrow$ Switch 2 controls the light
4. Bulb is off and cool $\rightarrow$ Switch 4 controls the light

The extension to four bulbs worked because we had two binary variables (illumination and bulb temperature) which creates 4 distinct states, of which we were only using 3 in the 3-bulb case.

Problem 2.4 [*Easy*]

How do you place 50 good candies and 50 rotten candies in two boxes such that if you choose a box at random and take out a candy at random, it better be good!

We need to maximize the probability of getting a good candy when selecting a random box and a random candy from it.

Solution. On one extreme, we can place all of the good candies in one box with all of the bad candies in another. This gives us $\mathbb{E}[G] = 0.5$. Thus, we should find solutions that better than 50%. We can beat this rather simply, by placing one good candy in one box and the other 49 in the other box with all of the bad candies. This results in:

$$\mathbb{E}[G] = 0.5 * \frac{1}{1} + 0.5 * \frac{49}{99} \approx 0.747$$

3 Code

3.1 Paradigms and Notes

Quick reference before the practice log. Fill this out as patterns recur.

- **Hashing:** trade space for $O(1)$ lookup. First reach for it on “find a pair / count occurrences / seen-before” problems.
- **Two pointers:** sorted array or linked list, converging or fast/slow. Turns many $O(n^2)$ scans into $O(n)$.
- **Sliding window:** contiguous subarray/substring under a constraint. Expand right, shrink left.
- **Binary search on the answer:** monotone feasibility predicate over a numeric range.
- **SQL:** window functions (ROW_NUMBER, RANK, SUM() OVER) and CTEs cover most DataLemur mediums.

3.2 Practice log

Two Sum (LeetCode 1). Return indices of the two numbers summing to a target.

Approach. One pass with a hash map from value to index. For each element check whether its complement was already seen.

Complexity: time $O(n)$, space $O(n)$.

```
1 def two_sum(nums, target):
2     seen = {}                                # value -> index
3     for i, x in enumerate(nums):
4         if target - x in seen:
5             return [seen[target - x], i]
6         seen[x] = i
7     return []
```

Listing 1: Two Sum, single-pass hash map

Second highest salary (DataLemur). Return the second-highest distinct salary, or NULL if none.

```
1 SELECT MAX(salary) AS second_highest
2 FROM (
3     SELECT salary,
4            DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
5     FROM employees
6 ) ranked
7 WHERE rnk = 2;
```

Listing 2: Second highest salary via DENSE_RANK

Style Reference Page

Definition 3.1 (Expectation). For a discrete random variable X with support $\mathcal{X}$, $\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \mathbb{P}(X = x)$, provided the sum converges absolutely.

Linearity of expectation

For any random variables $X_1, \dots, X_n$ and scalars a_i ,

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

This holds with no independence assumption. It is the single highest-yield identity for interview brainteasers, especially when combined with indicator variables $\mathbf{1}\{\cdot\}$.

Theorem 3.2 (Variance of a sum). *If $X_1, \dots, X_n$ are pairwise uncorrelated, then $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i)$.*

Proof. Expand $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$. Pairwise uncorrelatedness kills every cross term, leaving the claim. $\square$

Example 3.3. Draw a uniformly random permutation of $\{1, \dots, n\}$. Let X count fixed points. Write $X = \sum_i \mathbf{1}\{\pi(i) = i\}$. Each indicator has mean $1/n$, so by linearity $\mathbb{E}[X] = 1$, independent of n .

Add the variance derivation and the derangement limit $\mathbb{P}(X = 0) \rightarrow e^{-1}$.

Remark. Keep a running list of distributions with their mean, variance, and MGF. Interviewers probe recall speed on Bernoulli, Binomial, Geometric, Poisson, Exponential, and Normal.